CSE 4113
Introduction to Data Science
Chapter 6: Data Preparation

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Outlines

- Data Wrangling
- Data Cleaning
- Data Preprocessing

Data Wrangling

Data wrangling is the process of:

- transforming,
- organizing,
- restructuring,
- and enriching raw data

so that it becomes useful for analysis and machine learning.

Data Wrangling

Raw data collected from:

- websites,
- surveys,
- hospitals,
- banks,
- social media

is often difficult to analyze directly.

Data wrangling helps make it usable.

Why Data Wrangling

Problems in Raw Data

- Different formats
- Missing fields
- Multiple tables
- Repeated information
- Difficult structure

Without data wrangling:

- analysis becomes difficult,
- visualization becomes confusing,
- machine learning gives poor results.

Data Wrangling



Common Data Wrangling Operations

1. **Filtering:** Selecting important rows
2. **Merging:** Combining datasets
3. **Reshaping:** Changing table structure
4. **Aggregation:** Summarizing data
5. **Feature Creation:** Creating new columns

Filtering

Dataset:

Student	Department	CGPA
Sudipto	CSE	3.80
Ashraf	EEE	3.60
Maliha	CSE	3.95

Task: Show only CSE students.

Filtering

After Filtering:

Student	Department	CGPA
Sudipto	CSE	3.80
Maliha	CSE	3.95

Filtering removes unnecessary rows based on conditions.

Merging

Dataset 1:

ID	Name
1	Hasan
2	Kona

Dataset 2:

ID	CGPA
1	3.80
2	3.95

Task: Combine these datasets.

Merging

After Merging:

ID	Name	CGPA
1	Hasan	3.80
2	Kona	3.95

The two tables are combined using the common column.

Reshaping

Raw Dataset:

Student	Subject	Marks
A	Math	80
A	DS	85
B	Math	70
B	DS	90

Task: Change the structure or layout of data.

Reshaping

After reshaping:

Student	Math	DS
A	80	85
B	70	90

The data changed from long format → wide format

This makes analysis easier.

Aggregation

Dataset:

Department	Salary
CSE	50000
CSE	60000
EEE	55000
EEE	65000

Task: Find average salary by department.

Feature Creation

Dataset:

Student	Attendance (%)	Assignment	Final Exam
Hasan	90	18	70
Ayesha	75	15	60
Rahim	60	12	50

Create new useful features for analysis.

Feature Creation

New Feature: Total Marks

Formula: Total Marks=(attendance/10) + Assignment+Final Exam

Student	Attendance (%)	Assignment	Final Exam	Total Marks
Hasan	90	18	60	87
Ayesha	80	15	50	73
Rahim	60	12	40	58

Feature Creation

Performance Analysis:

Rule

Total Marks	Category
80+	Excellent
60–79	Good
Below 60	Weak

Performance

Student	Attendance (%)	Assignment	Final Exam	Total Marks
Hasan	90	18	60	87
Ayesha	80	15	50	73
Rahim	60	12	40	58

Data Cleaning

Data cleaning is the process of:

- detecting,
- correcting,
- removing

incorrect, incomplete, or inconsistent data.

Common Data Quality Problem

Problem	Example
Missing Values	Age = NULL
Duplicate Data	Same record repeated
Inconsistent Format	“Dhaka” vs “DHAKA”
Incorrect Data	Age = -5
Outliers	Salary = 10 কোটি

Missing Value

Original Data:

Student	Age	CGPA
Priya	22	3.80
Rakib	NULL	3.95
Munna	23	3.60

Handling Missing Value

Solution 1: Remove row.

Student	Age	CGPA
Priya	22	3.80
Munna	23	3.60

Solution 2: Fill Missing value

Student	Age	CGPA
Priya	22	3.80
Rakib	22	3.95
Munna	23	3.60

Filling Missing Value

1. Fill with Mean (Average)
Used for numerical data

Mean= $\sum x / n$

Example
Known ages are 22 and 23
Average Age=(22+23) / 2= 22.5
So, Age of Rakib will be filled by 22.5

Filling Missing Value

2. Fill with Median

Why Median?

Median works better when data contains outliers.

Dataset

Employee	Salary
Sagor	30000
Moin	32000
Maliha	NULL
Amit	35000
Raj	5000000

Filling Missing Value

Median Value

Sorted data:
30000, 32000, 35000, 5000000

Median:

Median=(32000+35000) / 2=33500

Employee	Salary
Sagor	30000
Moin	32000
Maliha	33500
Amit	35000
Raj	5000000

Instead of using a very large average affected by the outlier, we use the middle value.

Data Cleaning

3. Fill with Mode

Used for:

Categorical/Text Data

Here most frequent value is **Khulna**.

So, Amit's home town will be filled by **Khulna**.

Employee	Home Town
Sagor	Jashore
Moin	Dhaka
Maliha	Khulna
Amit	NULL
Raj	Khulna

Data Cleaning

4. Forward fill: Use previous value to fill missing data.
5. Backward fill: Use next value to fill missing data.
6. Use machine learning or statistical models to estimate missing values.

Duplicate Data

ID	Name	CGPA
1	Salman	3.80
2	Kona	3.95
2	Kona	3.95

Solution is simple. Just remove the duplicate row.

Duplicate records can distort statistics, increase bias, and affect machine learning.

Inconsistent Format

Khulna
khulna
KHULNA



Khulna
Khulna
Khulna

Pick one standard format. Standardization ensures consistency in data representation.

Incorrect Data

ID	Name	CGPA
1	Salman	-3.88
2	Kona	3.95
3	Atiq	378

Negative CGPA is invalid and 378 is also invalid as CGPA.

Incorrect or impossible values must be corrected or removed.

Outliers

An outlier is a value that is:

- very different,
- unusually high,
- or unusually low

compared to other data points.

Outliers

Observation

Most salaries are around 30,000–35,000

But **5,000,000** is extremely unusual.

This is an **Outlier**

Employee	Salary
Sagor	30000
Moin	32000
Maliha	35000
Amit	34000
Raj	5000000

Outliers

Why Outliers are Problematic

Outliers can:

- distort averages,
- affect machine learning models,
- create misleading graphs,
- reduce prediction accuracy.

Handling Outliers

1. Remove the Outlier

Delete rows which has clearly incorrect or impossible values.

When to Use

- data entry mistake,
- impossible value,
- measurement error.

Handling Outliers

2. Replace with Mean or Median

Replace extreme value with a more reasonable value.

Why?

Mean and Median are less affected by extreme values.

Handling Outliers

3. Capping (Winsorization)

Limit extreme values to a maximum or minimum threshold.

Suppose, Maximum acceptable salary = 100000. Replace all outliers greater than 100000 with 100000.

Benefit: Keeps the row while reducing extreme impact.

Handling Outliers

4. Transformation

Reduce large differences using mathematical transformation.

A common method: **Log Transformation**

Instead of *5000000* Use: *log(5000000)*

Large values become smaller and more manageable.

Detecting Outliers

1. IQR Method

Data: 2, 10, 12, 14, 15, 18, 20, 22, 100 (If not sorted, Sort it first)

Step 1: Find Median (Q2)

There are $n = 9$ values.

Median position:

$$\frac{n+1}{2} = \frac{9+1}{2} = 5$$

So the 5th value is the median.

$$Q_2 = 15$$

Step 2: Find Q1 (First Quartile)

Lower half (excluding median):

$$2,\ 10,\ 12,\ 14$$

Q1 is the median of this lower half.

Since there are 4 values:

$$Q_1 = \frac{10+12}{2}$$

$$Q_1 = \frac{22}{2} = 11$$

Detecting Outliers

Step 3: Find Q3 (Third Quartile)

Upper half (excluding median):

$$18,\ 20,\ 22,\ 100$$

Q3 is the median of this upper half.

$$Q_3 = \frac{20 + 22}{2}$$

$$Q_3 = \frac{42}{2} = 21$$

Step 4: Calculate IQR

$$IQR = Q_3 - Q_1$$

$$IQR = 21 - 11 = 10$$

Step 5: Find Lower and Upper Bound

Upper Bound

$$Q_1 - 1.5 \times IQR$$

$$11 - 1.5(10)$$

$$11 - 15 = -4$$

Lower Bound

$$Q_3 + 1.5 \times IQR$$

$$21 + 15 = 36$$

Detecting Outliers

Any value less than −4 greater than 36 is an outlier.

Dataset: 2, 10, 12, 14, 15, 18, 20, 22, 100

Only 100 > 36. So, 100 is an outlier.

Detecting Outliers

Step 1: Find Mean (μ)

Formula:

$$\mu = \frac{\sum x}{n}$$

Sum of all values:

$$2 + 10 + 12 + 14 + 15 + 18 + 20 + 22 + 100 = 213$$

Number of values:

$$n = 9$$

Mean:

$$\mu = \frac{213}{9}$$

$$\mu = 23.67$$

Detecting Outliers

Step 2: Find Standard Deviation (σ)

Formula:

$$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{n}}$$

Now calculate $(x-\mu)^2$:

Sum:

$$\sum(x-\mu)^2 = 6836.01$$

Variance:

$$\frac{6836.01}{9} = 759.56$$

Standard deviation:

$$\sigma = \sqrt{759.56}$$

$$\sigma \approx 27.56$$

x	(x-μ)	$(x-\mu)^2$
2	(2-23.67=-21.67)	469.59
10	(-13.67)	186.87
12	(-11.67)	136.19
14	(-9.67)	93.51
15	(-8.67)	75.17
18	(-5.67)	32.15
20	(-3.67)	13.47
22	(-1.67)	2.79
100	(76.33)	5826.27

Detecting Outliers

Formula: $z = \frac{x-\mu}{\sigma}$

For 100:

$$z = \frac{100 - 23.67}{27.56}$$

$$z = \frac{76.33}{27.56}$$

$$z \approx 2.77$$

Commonly, If |z|>3, then it is considered an outlier.

Here: |2.77|<3

So according to the standard Z-score rule 100 is not an outlier.

Data Preprocessing

Data preprocessing is the process of:

- transforming,
- encoding,
- scaling,
- and organizing data

before applying machine learning or analysis.

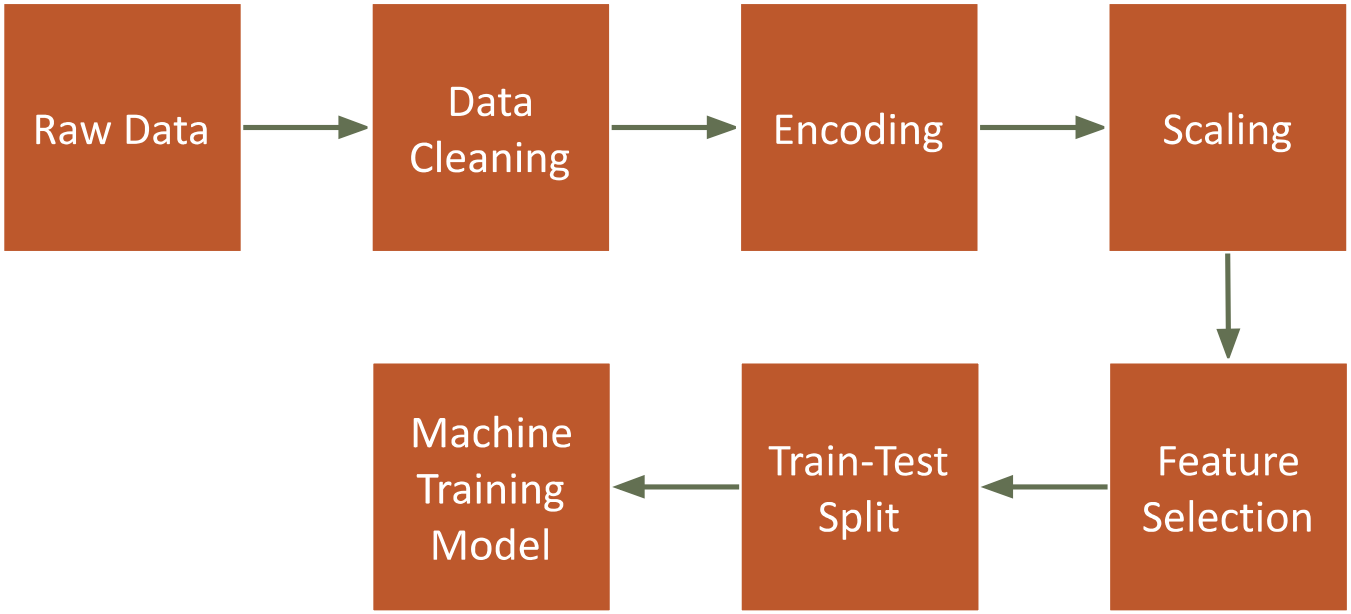
Why Data Preprocessing?

Machine Learning Models Need Proper Input

Raw or unprocessed data may:

- confuse algorithms,
- reduce accuracy,
- slow down training,
- produce biased results.

Common Data Preprocessing Tasks



Data Cleaning

Slide no. 18 to 42

Encoding

Machine learning models cannot directly understand text.

Like:

Gender
Male
Female
Other
Female

Label Encoding

After Encoding:

Gender	Encoded
Male	0
Female	1
Other	2

One-Hot Encoding

Problem with Label Encoding

The model may think:
1 > 0

which creates false order.

After One-Hot Encoding

Gender	Male	Female	Other
Male	1	0	0
Female	0	1	0
Other	0	0	1

Feature Scaling

Different features may have very different ranges.

Salary values are much larger than Age values.

Machine learning models may give too much importance to salary.

Age	Salary
22	30000
25	50000
30	100000

Feature Scaling

1. Min-Max Scaling

Formula:

$$X' = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

Feature Scaling

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

A. Min-Max Scaling for Age

- Minimum age = 22
- Maximum age = 30

$$\text{Range} = 30 - 22 = 8$$

For Age = 22

$$\frac{22 - 22}{8} = 0$$

For Age = 25

$$\frac{25 - 22}{8} = \frac{3}{8} = 0.375$$

For Age = 30

$$\frac{30 - 22}{8} = \frac{8}{8} = 1$$

Feature Scaling

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

B. Min-Max Scaling for Salary

- Minimum salary = 30000
- Maximum salary = 100000

$$\text{Range} = 100000 - 30000 = 70000$$

For Salary = 30000

$$\frac{30000 - 30000}{70000} = 0$$

For Salary = 50000

$$\frac{50000 - 30000}{70000} = \frac{20000}{70000} = 0.286$$

For Salary = 100000

$$\frac{100000 - 30000}{70000} = 1$$

Feature Scaling

Min-Max Scaled Dataset

Age	Salary
0.000	0.000
0.375	0.286
1.000	1.000

Feature Scaling

2. Standardization (Z-score Normalization)

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- μ = Mean
- σ = Standard Deviation

Feature Scaling

A. Standardization for Age

Ages: 22, 25, 30

Step 1: Mean

$$\mu = \frac{22 + 25 + 30}{3} = \frac{77}{3} = 25.67$$

Step 2: Standard Deviation

$$\sigma = \sqrt{\frac{(22 - 25.67)^2 + (25 - 25.67)^2 + (30 - 25.67)^2}{3}}$$

$$= \sqrt{\frac{13.47 + 0.45 + 18.75}{3}}$$

$$= \sqrt{10.89} = 3.30$$

Step 3: Z-scores

For 22

$$\frac{22 - 25.67}{3.30} = -1.11$$

For 25

$$\frac{25 - 25.67}{3.30} = -0.20$$

For 30

$$\frac{30 - 25.67}{3.30} = 1.31$$

Feature Scaling

B. Standardization for Salary

Salaries: 30000, 50000, 100000

Step 1: Mean

$$\mu = \frac{30000 + 50000 + 100000}{3} = 60000$$

Step 2: Standard Deviation

$$\sigma = \sqrt{\frac{(30000 - 60000)^2 + (50000 - 60000)^2 + (100000 - 60000)^2}{3}}$$

$$= \sqrt{\frac{900000000 + 100000000 + 1600000000}{3}}$$

$$= \sqrt{866666666.67} \approx 29439$$

Step 3: Z-scores

For 30000

$$\frac{30000 - 60000}{29439} = -1.02$$

For 50000

$$\frac{50000 - 60000}{29439} = -0.34$$

For 100000

$$\frac{100000 - 60000}{29439} = 1.36$$

Feature Scaling

Standardized Dataset

Age	Salary
-1.11	-1.02
-0.20	-0.34
1.31	1.36

Feature Selection

Feature selection means:

choosing the most important features (columns)

and removing:

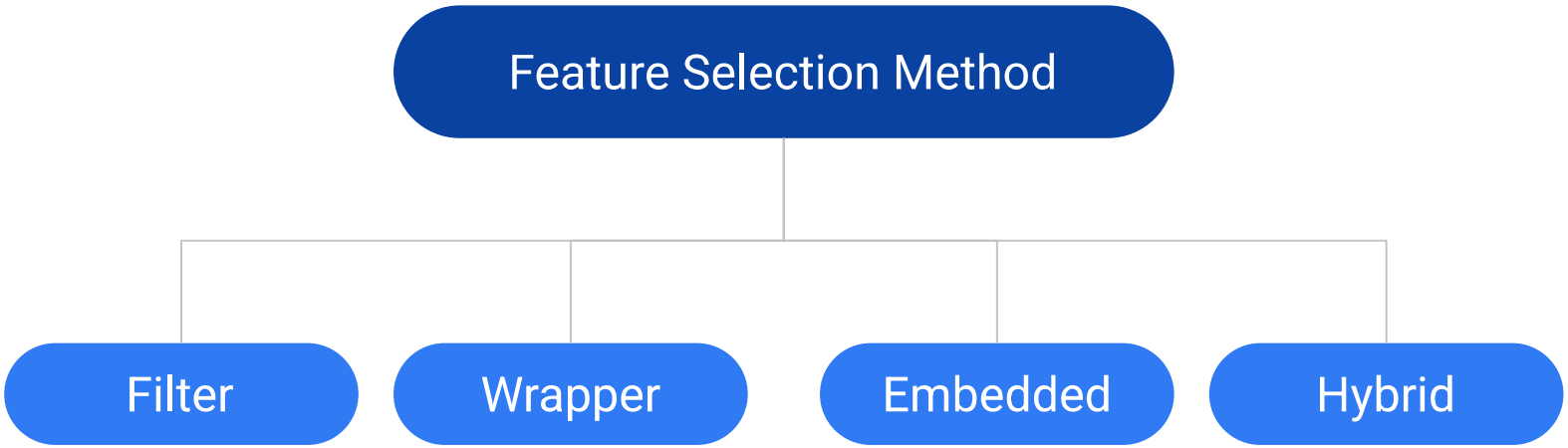
- irrelevant,
- redundant,
- noisy features.

Feature Selection

Why Important?

- Improves model accuracy
- Reduces training time
- Prevents overfitting
- Makes models simpler

Feature Selection Methods



Filter Method

Evaluate features independently using statistics correlation, information measures.

Characteristics

Fast, Scalable, Ignores model interaction

Filter Method

Method	Purpose
Variance Threshold	Remove near-constant features
Pearson Correlation	Measure linear relationship
ANOVA F-test	Compare group variance
Chi-Square Test	Categorical feature selection
Mutual Information	Detect linear & nonlinear dependency

Wrapper Method

Select features by **training and evaluating a model repeatedly.**

Characteristics

More accurate, Computationally expensive

Wrapper Method

Method	How it Works
Forward Selection	Add best features one by one
Backward Elimination	Remove worst features one by one
RFE	Recursively remove least important features
Exhaustive Search	Test all feature combinations

Embedded Method

Feature selection happens **during model training.**

Characteristics

Balance of speed and accuracy, Built into learning algorithms

Embedded Method

Method	Associated Model
LASSO (L1)	Linear/Logistic Regression
Ridge (L2)	Regression
Elastic Net	Combined regularization
Tree Importance	Random Forest, XGBoost
Permutation Importance	Any model

Hybrid Method

Combine multiple feature selection strategies.

Method	Best Use
Boruta	High-dimensional datasets
SHAP-based Selection	Explainable AI
Genetic Algorithms	Large search spaces
mRMR	Genomics, bioinformatics
CFS	Correlation-based selection

Quick Decision Guide

Scenario	Recommended Method
Very large dataset	Variance Threshold + Mutual Information
High accuracy needed	RFE
Interpretable model	LASSO
Thousands of features	Chi-Square, Boruta
Nonlinear relationship	Mutual Information
Black-box model	Permutation Importance

Dimensionality Reduction(idea)

Problem
Too many features can:

- slow models,
- increase complexity,
- cause overfitting.

Solution
Reduce unnecessary features while preserving important information.

Example Techniques

- PCA (Principal Component Analysis)
- Feature Selection

Train-Test Split

Split dataset into:

- training data
- testing data

To check whether the model works well on unseen data.

Typical Split

Training Data* → 80%

Testing Data* → 20%

***The ratio of training and testing data may vary on demand.**

Thank you

for your Attention